VSB TECHNICAL UNIVERSITY OF OSTRAVA

Video Quality Assessment

440-2216/01 Quality of Service · Lecture 06

Jan Rozhon · Department of Telecommunications, FEECS

Where we are going

Part 1

Digital video and where quality is lost

Pixels, RGB and YCbCr, raw vs compressed size, codecs and containers, hybrid coding, artefacts

↔ what the metrics have to detect

Part 2

Subjective assessment

ITU-T P.910, ITU-R BT.500: viewers, content, methods, scales

↔ the ground truth, as MOS was for speech

Part 3

Objective assessment

PSNR, SSIM, VMAF; no-reference and parametric models

↔ Lab 04 implements PSNR and SSIM

One question runs through the lecture: **which property of the picture does a number measure — and how closely does it follow what viewers actually see?**

What you should be able to do

1. Describe how a digital video is represented — resolution, frame rate, bit depth, RGB vs Y′CbCr, chroma subsampling — compute its raw bit rate, and tell a codec from a container.
2. Explain where in the delivery chain video quality is lost, why lossy compression is unavoidable, and how a single packet loss propagates through a group of pictures.
3. Design a subjective video test according to ITU-T P.910 / ITU-R BT.500: viewers, content selection by SI/TI, method (ACR, ACR-HR, DCR, PC, DSCQS) and rating scale.
4. Compute MSE and PSNR, explain the luminance, contrast and structure terms of SSIM, and explain why the two can disagree.
5. Describe how VMAF fuses elementary features, how metrics are validated against subjective scores, and when a no-reference or parametric model is the right tool.

Keep asking: **full reference, reduced reference, no reference or no pixels at all — and validated on which content?**

A video metric is judged by one criterion only

How well it predicts what viewers would say — measured against a subjective test

Part 1

Digital video and where its quality is lost

What a digital video is

A **frame** is a two-dimensional array of **pixels** (picture elements); a video is a sequence of frames. Four numbers describe its raw form:

- **resolution** $W \times H$ — pixels per frame, e.g. 1920 × 1080
- **frame rate** f — frames per second: 24 (cinema), 25 / 50 (Europe), 30 / 60 (Americas, games)
- **bit depth** b — bits per sample: 8 bit = 256 levels (SDR), 10 bit = 1024 levels (HDR); too few levels show as banding
- **colour format** — every pixel needs **three samples**, one per colour component (next slides)

The *p* in 1080p means **progressive**: every frame holds all lines. The older *i* (1080i) is **interlaced**: odd and even lines are sent as alternate half-frames (fields), a legacy of analogue television.



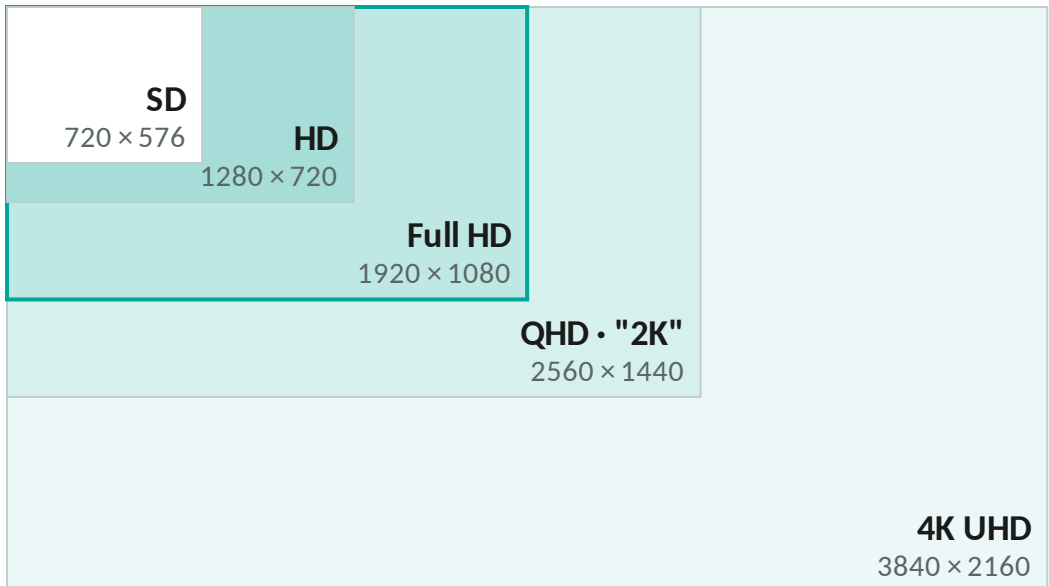


The marked 24 × 14 px area, enlarged 16 ×: each square is one pixel.

Pink square, bottom centre: (R, G, B) = (159, 107, 128).

Resolutions — from SD to 4K

TO SCALE



PIXELS PER FRAME

576p	SD	0.41 M	0.2 ×
720p	HD	0.92 M	0.44 ×
1080p	Full HD	2.07 M	1 ×
1440p	QHD · "2K"	3.69 M	1.8 ×
2160p	4K UHD	8.29 M	**4 ×**

last column: relative to Full HD

Cinema (DCI) uses wider frames:
2K = 2048 × 1080, 4K = 4096 × 2160.
Consumer "2K" usually means 1440p.

Doubling the resolution in each direction quadruples the pixels — and the raw bit rate.

The names mix two conventions: **1080p** counts lines, **4K** counts (roughly) columns. More pixels do not mean more quality by themselves — a 720p stream upscaled on a 4K television has 4K pixels and 720p detail, and the viewing distance decides whether the extra pixels are visible at all (Part 2).

Colour models — RGB and Y′CbCr

RGB — how cameras and displays work

- additive mixing of red, green and blue primaries, matching the eye's three types of cones
- 3 × 8 bit = **24 bit/pixel**, 16.7 million colours
- the three components are **strongly correlated**: each repeats the full detail of the picture (next slide), so coding them separately wastes bits

Y′CbCr — how video is stored and coded

- **luma** Y' (brightness) separated from two **chroma** colour differences (ITU-R BT.709):

$$Y' = 0.2126\,R' + 0.7152\,G' + 0.0722\,B'$$

$$C_B = \frac{B' - Y'}{1.8556}, \qquad C_R = \frac{R' - Y'}{1.5748}$$

- green dominates luma: the eye is most sensitive to it
- a 3 × 3 matrix — reversible up to rounding

YUV is the analogue-TV name of the same idea; digital video uses Y′CbCr, but software says YUV for both (ffmpeg's `yuv420p`). The prime marks gamma-corrected values. Separating brightness from colour is what let colour television stay compatible with black-and-white receivers in the 1950s.

The same picture in RGB and in Y′CbCr



Original · colour



R



G



B



Y′ · luma, full size



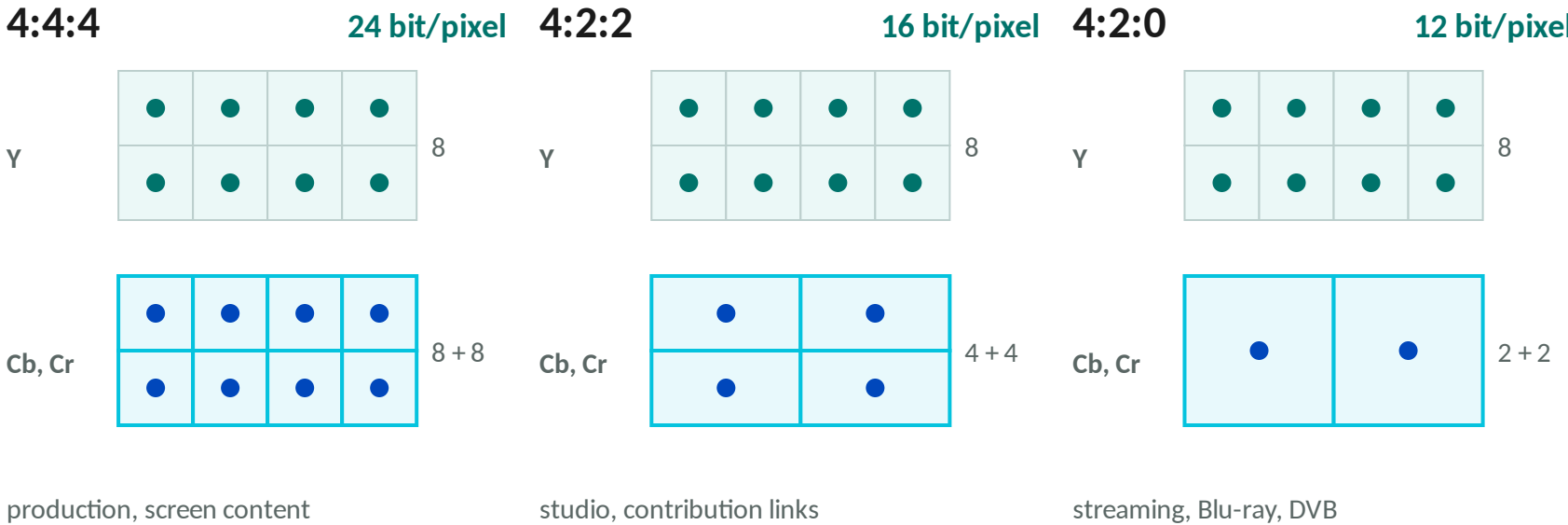
Cb · ¼ of the samples



Cr · ¼ of the samples

R, G and B look almost the same. In Y′CbCr the detail sits in **Y′**; the chroma planes carry little detail and little contrast (stretched 3 × here) — shown at their **true 4:2:0 size**.

Chroma subsampling



production, screen content

studio, contribution links

streaming, Blu-ray, DVB

Each coloured block shares one Cb and one Cr value; luma keeps full resolution because the eye resolves detail mainly in brightness.

J:a:b — in a block J pixels wide and two rows high, a chroma samples in the first row and b new ones in the second. Average bits per pixel with chroma subsampled by $s_x \times s_y$: $b_{\text{px}} = b\,(1 + 2/(s_x s_y))$, i.e. $8 \cdot 1.5 = 12$ bit for 4:2:0.

4:2:0 halves the raw data with practically no visible loss in natural video. It shows on sharp coloured edges — red text, graphics, game HUDs — as **colour bleeding**, which is why screen sharing and desktop streaming prefer 4:4:4.

How big is raw video?

$$R_{\text{raw}} = W \cdot H \cdot f \cdot b_{\text{px}} \qquad \text{e.g.} \qquad 1920 \cdot 1080 \cdot 30 \cdot 12 \approx 746\ \text{Mbit/s}$$

Format	Resolution	Raw · 30 fps, 8-bit 4:2:0	1 h raw	Typical stream	1 h streamed	Ratio
Full HD	1920 × 1080	746 Mbit/s	336 GB	5 Mbit/s · H.264	2.3 GB	**≈ 150 : 1**
QHD "2K"	2560 × 1440	1.33 Gbit/s	597 GB	≈ 9 Mbit/s · HEVC, VP9	4.1 GB	**≈ 150 : 1**
4K UHD	3840 × 2160	2.99 Gbit/s	1.34 TB	15 Mbit/s · HEVC, AV1	6.8 GB	**≈ 200 : 1**

A two-hour 4K film: **2.7 TB** raw, at most 100 GB on a UHD Blu-ray, **13.5 GB** streamed.

4K at 60 fps with 10-bit HDR: $3840 \cdot 2160 \cdot 60 \cdot 15 \approx 7.5$ Gbit/s raw — about a hundred times a typical home connection.

Checkpoint — a video call

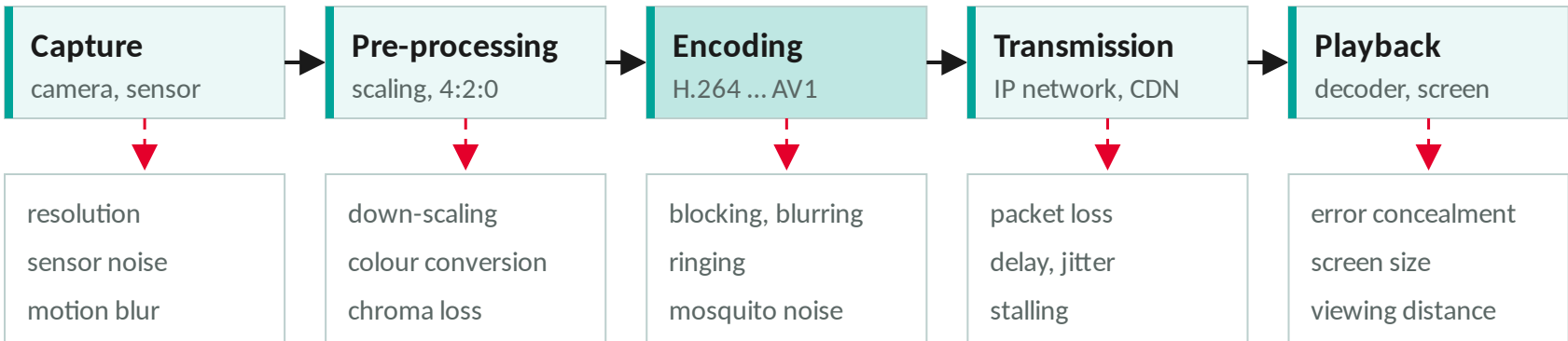
A video call sends 720p at 30 frames/s, 8-bit 4:2:0, in 1.5 Mbit/s. What compression ratio does the encoder achieve? And what happens to the ratio if the camera image switches to 4:4:4 at the same bit rate?

Predict first · discuss with a neighbour · then reveal

> $1280 \cdot 720 \cdot 30 \cdot 12 \approx 332$ Mbit/s raw, so about **220 : 1**. With 4:4:4 the raw rate doubles to 24 bit/pixel (≈ 664 Mbit/s) and the encoder must compress **≈ 440 : 1** in the same 1.5 Mbit/s — it can only do so by quantising more coarsely, so the picture gets worse even though it carries "more colour".

The video delivery chain

SOURCE → VIEWER



Signal quality is lost at every stage; what the viewer finally judges also depends on the display and on the viewing context — the QoS/QoE distinction of Lecture 03 applies to video as it did to speech.

Two evaluation approaches, as for speech: **subjective** — ask viewers; **objective** — compute a number from the signal or from parameters. Both must state *which* stages they cover: a codec comparison isolates encoding, a service-level test covers the whole chain.

Why compression is unavoidable

Raw 1080p30 needs 746 Mbit/s, a typical stream 5 Mbit/s: **about 150 : 1**. A codec gets there by removing four kinds of redundancy — and by discarding what viewers are unlikely to miss:

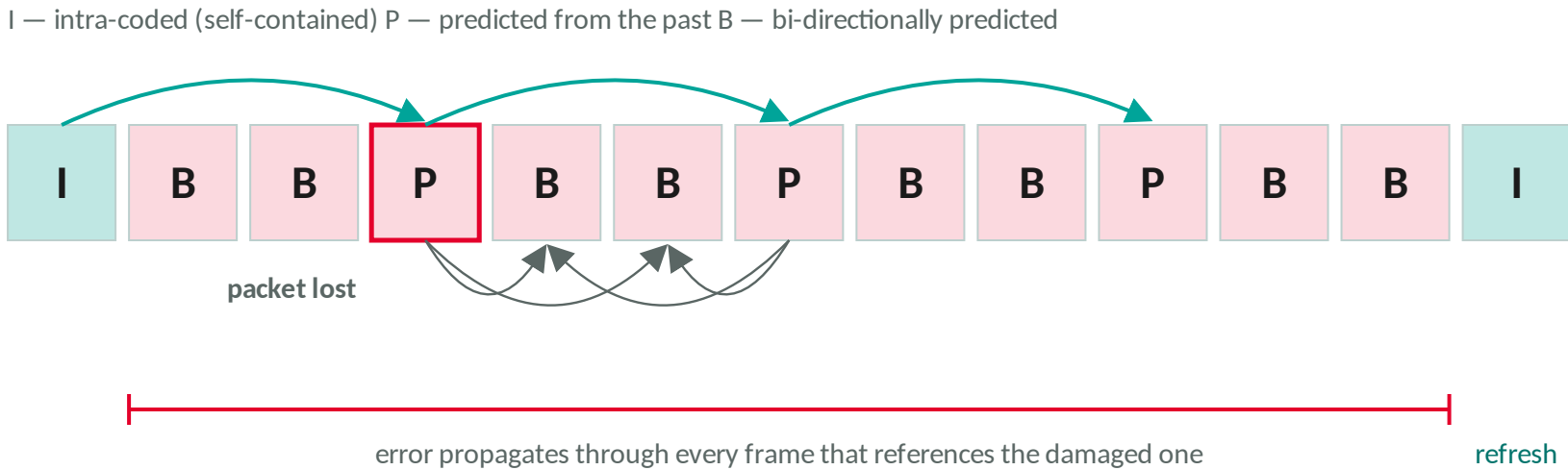
- **spatial** — neighbouring pixels are similar → intra prediction, transform
- **temporal** — consecutive frames are similar → inter prediction
- **psychovisual** — the eye misses fine detail and colour detail → quantisation, 4:2:0
- **statistical** — some values are far more frequent → entropy coding

Only the psychovisual part loses information — and it is exactly where quality is lost.

Hybrid video coding — common to every standard from H.261 to H.266 and AV1:

1. **Prediction** — *intra* from neighbouring blocks of the same frame, *inter* from other frames by motion compensation; only the residual is coded.
2. **Transform** — the residual block is converted to spatial frequencies (a DCT or its integer approximation).
3. **Quantisation** — coefficients are divided by a step size and rounded. **This is the only lossy step**, controlled by the quantisation parameter (QP).
4. **Entropy coding** — lossless (CAVLC, CABAC, arithmetic coding).

Groups of pictures and error propagation



Inter prediction is what makes compression efficient — and what makes video fragile. A lost packet damages not one frame but **every frame predicted from it** until the next I-frame (or intra refresh). The same loss rate therefore hurts video far more when it hits a reference frame than a non-reference B-frame.

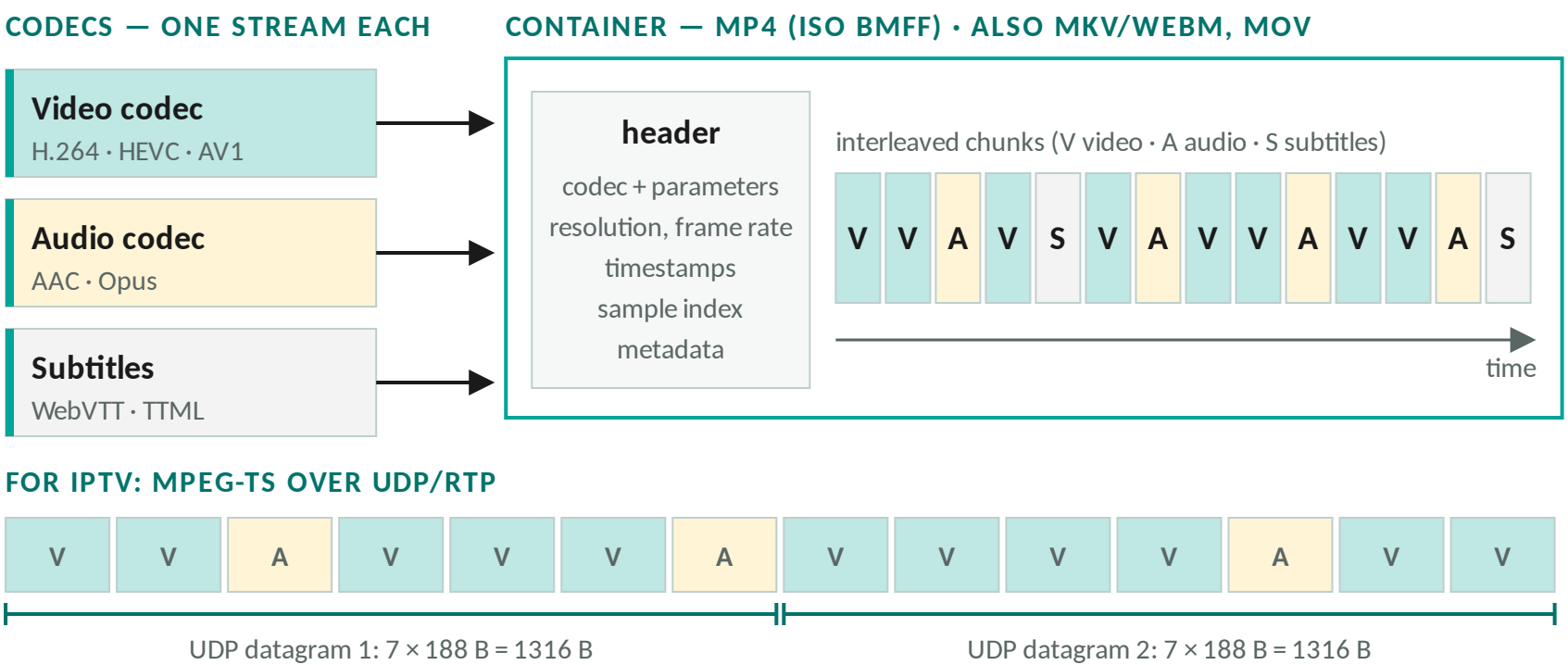
Compression standards

Generation	Standard	Year	Developed by	Typical use
—	MPEG-2 / H.262	1995	ISO/IEC MPEG + ITU-T	DVD, first-generation DVB-T
1	**H.264 / AVC** (MPEG-4 Part 10)	2003	JVT (MPEG + VCEG)	Blu-ray, web video, videoconferencing
2	**H.265 / HEVC** (MPEG-H Part 2)	2013	JCT-VC	UHD broadcasting, DVB-T2 in Czechia
3	**H.266 / VVC**	2020	JVET	UHD/8K, 360° video
2	**VP9**	2013	Google	YouTube, WebRTC
3	**AV1**	2018	Alliance for Open Media	royalty-free streaming (Netflix, YouTube)

Rule of thumb: each generation needs **up to about half the bit rate** of its predecessor for the same *subjective* quality, averaged over broad test sets — at the cost of much higher encoder complexity. On a single clip the saving can be far smaller (Part 3 measures it).

AV1 merged the unreleased VP10 (Google), Daala (Xiph.Org/Mozilla) and Thor (Cisco) into one royalty-free design; VP8 (2010) was VP9's predecessor.

Codec vs container



A **codec** (coder–decoder) compresses *one* stream — it decides the quality. A **container** packs several coded streams into one file or stream, with the timestamps that keep them in sync — it decides how the streams travel, not how they look.

Containers in practice

Container	Typical content	Used for
MP4 (ISO BMFF), .mp4 / .m4s	H.264, HEVC, AV1 · AAC	files; DASH/HLS segments (CMAF) — Lecture 07
Matroska / WebM, .mkv / .webm	anything · WebM: VP9, AV1, Opus	files, browser video
MPEG-TS, .ts	H.264, HEVC · AAC, AC-3	DVB broadcast, IPTV, older HLS
RTP — a protocol, not a file	one stream per RTP session	video calls, WebRTC — Lecture 05

The file name tells you the **container only** — ask the file itself:

```
$ ffprobe reference.mp4
Input #0, mov,mp4,m4a,3gp,3g2,mj2, ...
  Duration: 00:00:02.00, bitrate: 9919 kb/s
  Stream #0:0: Video: h264 (High),
    yuv420p(progressive), 854x480, 50 fps
```

Container *mp4*, codec *H.264*, Y′CbCr 4:2:0, progressive — everything from this part in four lines.

Typical impairments

Compression artefacts — spatial

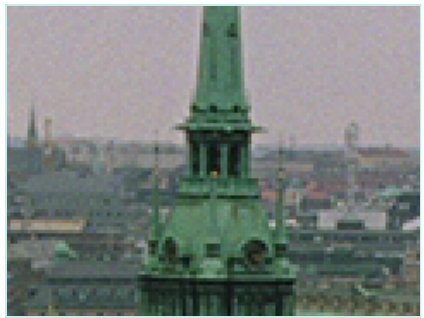
- **blocking** — visible edges of transform blocks at coarse quantisation
- **blurring** — high-frequency coefficients quantised to zero
- **ringing, mosquito noise** — oscillations around sharp edges, flickering in time
- **colour bleeding** — a consequence of chroma subsampling
- **banding** — steps in smooth gradients (sky)

Temporal and transmission impairments

- **jerkiness** — reduced frame rate or dropped frames
- **smearing, frozen blocks** — packet loss plus error concealment
- **freezing** — decoder waits for a lost reference frame
- **stalling (rebuffering)** — the player's buffer runs empty in HTTP adaptive streaming
- **quality switching** — adaptive bit rate changes resolution mid-stream

A metric can only react to what it can see: a per-frame image metric has no notion of stalling, and a stalling counter has no notion of blocking.

What compression artefacts look like



Reference · uncompressed



Blocking · MPEG-2, 1.5 Mbit/s



Blurring · H.264, 250 kbit/s



Ringing, mosquito noise · HEVC, 150 kbit/s

Each codec fails in its own way. **MPEG-2** shows its 8 × 8 transform blocks. **H.264** has an in-loop **deblocking filter**, which trades the block edges for blur. **HEVC** keeps edges sharper with larger transforms, but the quantisation error oscillates around them. A metric has to catch all three.

Crops of 120 × 90 px, enlarged 2 ×, from the SVT test sequence *old_town_cross* (720p50, no copyright), 2 s encodes with ffmpeg.

Impairments in motion



Reference — park_joy, 720p50



MPEG-2 at 3 Mbit/s — blocking that moves with the scene



H.264, 3 datagrams lost in frame 12 — smearing until the I-frame at 1 s



Stall — the picture freezes for 1.2 s, then resumes

Click a clip to play it (loops, no sound). Watch the **packet-loss** clip closely: the damaged area is dragged along with the motion for almost a second — the error propagation of the GOP slide made visible. The **stall** loses no pixel at all, yet viewers rate it among the worst impairments.

Checkpoint — same bit rate, same quality?

Two 1080p streams are encoded with the same encoder at the same bit rate, 4 Mbit/s. One is a studio interview, the other a football match filmed with a moving camera. Will they look equally good? What does that imply for a subjective test?

Predict first · discuss with a neighbour · then reveal

> **No.** The football match has far more motion and fine texture (grass, crowd), so prediction leaves larger residuals and the encoder must quantise more coarsely to hit the same bit rate — it shows artefacts the interview does not. Quality is a property of **content × condition**, not of the bit rate alone, so a test set must span a range of spatial and temporal complexity (SI and TI, next part) or its results will not generalise.

Part 2

Subjective assessment

Why ask viewers at all?

Properties of subjective assessment

- human observers rate what they see — the **most reliable** measure of perceived quality and the reference for every objective metric
- **time-consuming** and **costly**: laboratory, calibrated displays, tens of viewers, hours of material
- not repeatable on demand — every new codec setting means a new test

The result is influenced by (Qualinet QoE influence factors)

- **human** — individual interests, expectations, visual acuity, experience with the medium
- **system** — display size, resolution, calibration, tools and equipment
- **context** — ambient light, viewing distance, task (entertainment vs. surveillance)

Standards exist to fix as many of these factors as possible, so that results from different laboratories can be compared.

The standards

Recommendation	Scope	Methods
ITU-T P.910	multimedia video: videoconferencing, streaming, storage	ACR, ACR-HR, DCR, PC (+ SDSCE in an appendix)
ITU-R BT.500	television pictures, broadcast quality	DSIS, DSCQS, SSCQE, SDSCE, …
ITU-R BT.1788	multimedia, multi-stimulus with random access	SAMVIQ
ITU-T P.913	any environment: home, public places, crowdsourcing	ACR, ACR-HR, DCR, PC

ITU-T P.910 — *Subjective video quality assessment methods for multimedia applications*: first published 1996, revised 1999, 2008 and repeatedly since 2021; the current edition is from 2026.

The methods overlap under different names: P.910's **ACR** is BT.500's **single stimulus (SS)**; P.910's **DCR** is BT.500's **DSIS**.

Designing the test — viewers and viewing conditions

Viewers (P.910)

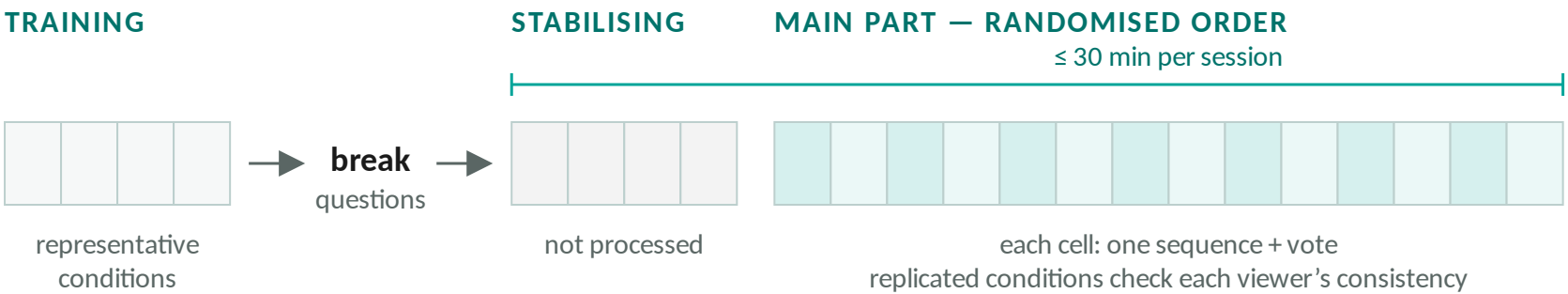
- from **4 to 40** depending on the required validity — 4 is the statistical minimum, beyond 40 there is rarely any benefit
- **at least 15** should normally participate; P.913 asks for ≥ 24 after screening, ≥ 35 in public environments
- **non-experts** — not involved in picture-quality work; experts (4–8) only for pilot tests
- screened for **visual acuity** (no errors on the 20/30 line of an eye chart) and **colour vision** (≤ 2 of 12 Ishihara plates missed)

Viewing conditions (P.910, Table 1)

Parameter	Setting
viewing distance	1–8 H (picture heights)
peak screen luminance	100–200 cd/m²
background room illumination	≤ 20 lx
background chromaticity	D65

Session length — at most about **30 minutes**, including training; longer tests are split into sessions with breaks.

Structure of a test session



Randomisation cancels order and fatigue effects; the stabilising items absorb the "first impressions".

Training explains the scale and shows the range of quality to expect, without telling viewers what "good" is. **Stabilising** sequences absorb the first, less consistent votes. **Replications** of identical conditions measure each viewer's consistency and allow unreliable viewers to be discarded.

Choosing the test content — SI and TI

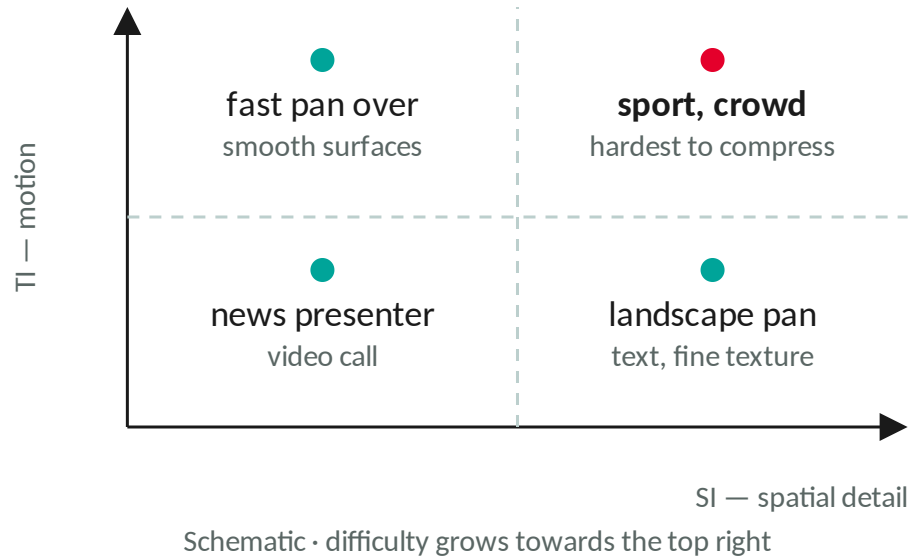
P.910 quantifies how demanding a sequence is:

$$\mathrm{SI} = \max_n \left\{ \mathrm{std}_{i,j} \left[\mathrm{Sobel}(F_n) \right] \right\}$$

$$M_n(i,j) = F_n(i,j) - F_{n-1}(i,j)$$

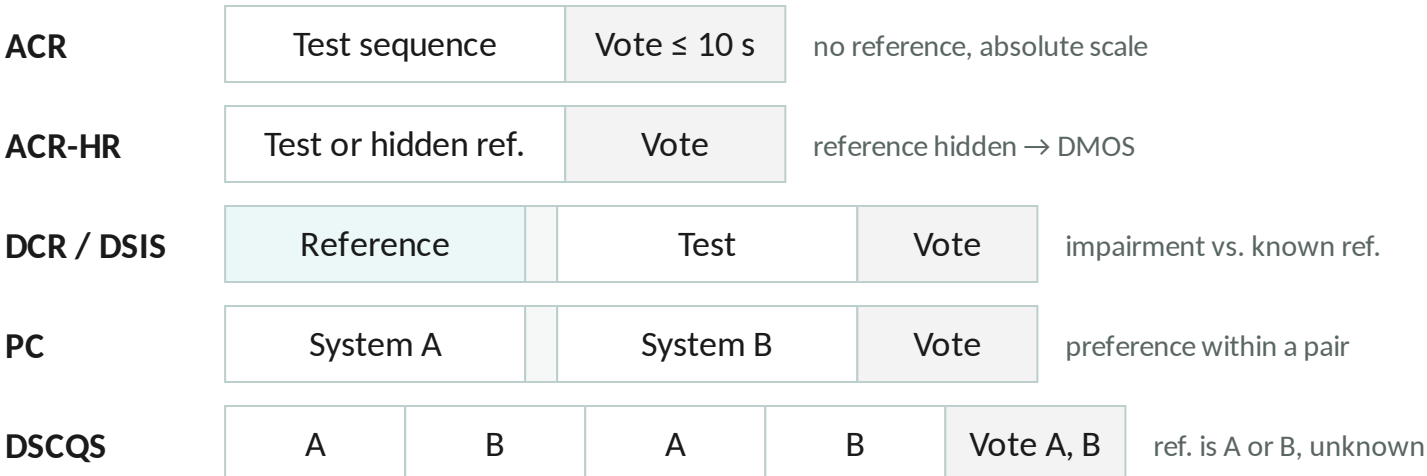
$$\mathrm{TI} = \max_n \left\{ \mathrm{std}_{i,j} \left[M_n(i,j) \right] \right\}$$

F_n is the luminance of frame n. **SI** grows with edges and texture, **TI** with motion. Test sequences (~10 s each) should cover all four quadrants of the SI–TI plane.



Schematic · difficulty grows towards the top right

Test methods — presentation patterns



~10 s per sequence · grey pause ≈ 2 s between the two stimuli of a pair (ITU-T P.910)

ACR is fast and resembles normal viewing. **DCR** is more sensitive to small impairments because the viewer sees the reference first. **PC** discriminates best between similar systems, but the number of pairs grows as $n(n-1)$.

The methods at a glance

Method	Reference	Scale	Output	Best for
ACR — Absolute Category Rating (= SS)	none	5-grade quality	MOS	many conditions, realistic viewing
ACR-HR — ACR with Hidden Reference	hidden	5-grade quality	DMOS	removing the influence of the source content
DCR — Degradation Category Rating (= DSIS)	explicit, first	5-grade impairment	DMOS	high-quality systems, small impairments
PC — Pair Comparison	the other system	preference	preference scale	ranking similar systems
DSCQS — Double Stimulus Continuous Quality Scale	unknown, A or B	continuous 0–100	difference score	broadcast codec evaluation
SSCQE — Single Stimulus Continuous Quality Evaluation	none	slider, sampled in time	quality over time	long clips with varying quality
SDSCE — Simultaneous Double Stimulus for Continuous Evaluation	side by side	slider, fidelity 0–100	fidelity over time	sparse transmission errors

Rating scales



More grades add resolution only if viewers can use it; verbal labels anchor the meaning of the numbers.

The **nine-grade** scale adds resolution where the five categories are too coarse — typically when comparing low-bit-rate codecs. The **eleven-grade** scale adds anchored endpoints. **Continuous** scales avoid forcing a vote into a category. All of them keep the verbal labels, because the labels give the numbers their meaning.

From votes to a result

For a condition rated by N viewers, exactly as in Lecture 05:

$$\mathrm{MOS} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\mathrm{CI}_{95\,\%} = \mathrm{MOS} \pm t_{0.975,\,N-1}\,\frac{s}{\sqrt{N}}$$

Report MOS **per condition**, averaged over contents, *and* per content — a codec may fail only on high-TI material.

ACR-HR scores each sequence against the hidden reference of the same content, then averages into a DMOS:

$$\mathrm{DV} = V(\mathrm{test}) - V(\mathrm{ref}) + 5$$

Before averaging

- discard the training and stabilising votes
- screen out inconsistent viewers (BT.500 Annex 1, correlation with the panel in P.913)
- keep the scale's name: MOS from ACR, DMOS from ACR-HR or DCR, difference scores from DSCQS are **not interchangeable**

What the result is used for

- ranking codecs or settings
- the **ground truth** for training and validating objective metrics — Part 3

Checkpoint — which method?

A streaming provider wants to know whether its new encoder at 3 Mbit/s is **indistinguishable** from the current one at 4 Mbit/s on a 4K television. Should the test use ACR or DCR — and why?

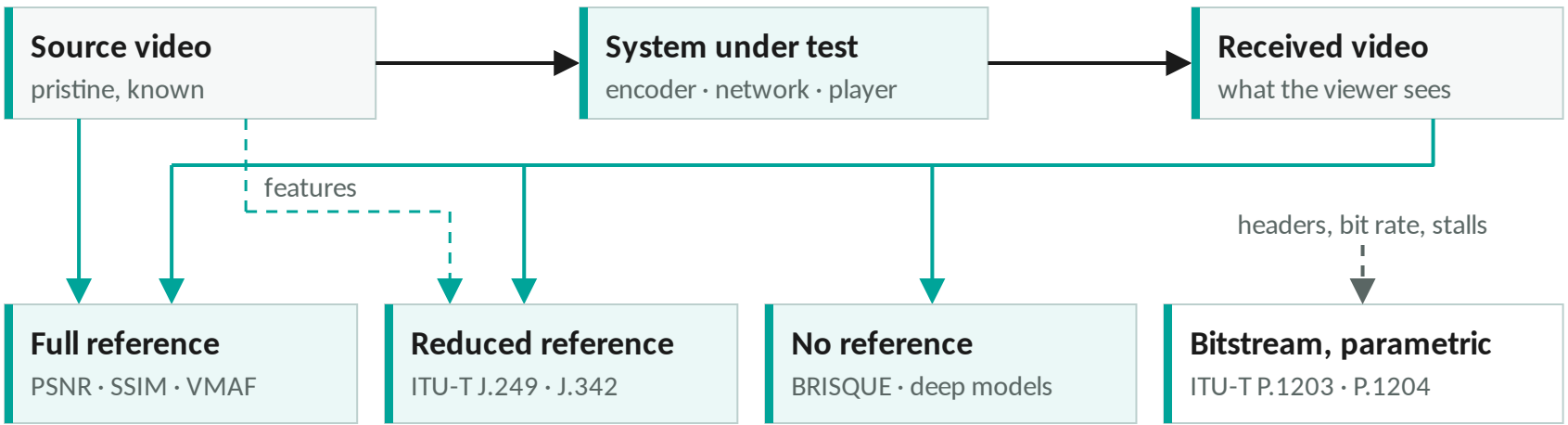
Predict first · discuss with a neighbour · then reveal

DCR (or a pair comparison). Both versions are close to transparent, so on the absolute ACR scale both land at "Good"–"Excellent" and the difference drowns in the scale context and the viewers' spread. DCR shows the source first and asks about the *impairment*, which separates "imperceptible" from "perceptible but not annoying". The display (4K TV) and viewing distance (≈ 1.5 H for UHD) must match the service, or the test measures a different product.

Part 3

Objective assessment

Four families of objective models



The same trade-off as for speech: less input → easier deployment → usually lower accuracy.

Full reference (FR) needs the pristine source, so it lives in the encoding lab and in codec development. **No reference (NR)** and **parametric** models are what a network operator can deploy on live traffic — the video counterparts of P.563 and the E-model.

MSE and PSNR

For a reference image I and a test image K of $m \times n$ pixels:

$$\mathrm{MSE} = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \left[I(i,j) - K(i,j) \right]^2$$

$$\begin{aligned} \mathrm{PSNR} &= 10 \log_{10} \frac{\mathrm{MAX}_I^2}{\mathrm{MSE}} = 20 \log_{10} \frac{\mathrm{MAX}_I}{\sqrt{\mathrm{MSE}}} \\ &= 20 \log_{10} \mathrm{MAX}_I - 10 \log_{10} \mathrm{MSE} \end{aligned}$$

MAX_I is the maximum **possible** pixel value — 255 for 8-bit video, 1023 for 10-bit — not the largest value in the image.

Example: 8-bit frames, MSE = 100:

$$\mathrm{PSNR} = 10 \log_{10} \frac{255^2}{100} \approx 28.1 \text{ dB}$$

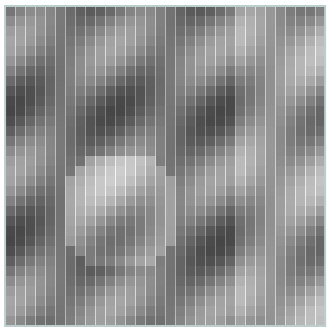
For video

- usually computed on luminance (Y) only, or as a weighted Y:U:V average (e.g. 6:1:1)
- per frame, then **pooled** over time — the mean of frame PSNRs differs from the PSNR of the mean MSE, because the logarithm is nonlinear
- identical frames give MSE = 0 and PSNR = ∞; tools report ∞ or cap it at a fixed value

There is no universal "good" PSNR threshold: the same value can look fine on one content and poor on another.

Equal MSE, very different pictures

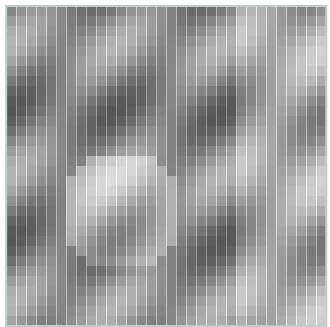
Reference



MSE 0 · PSNR ∞

SSIM 1.00

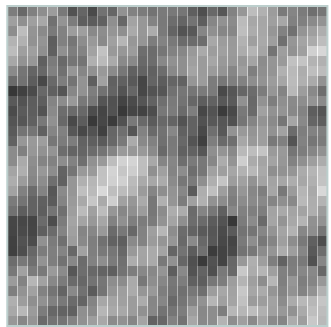
Mean shift



MSE 225 · PSNR 24.6 dB

SSIM 0.99

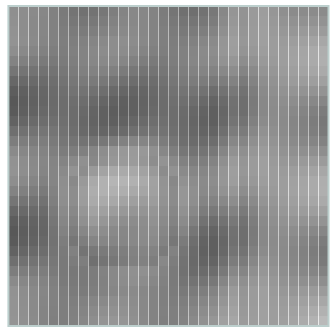
Gaussian noise



MSE 225 · PSNR 24.6 dB

SSIM 0.84

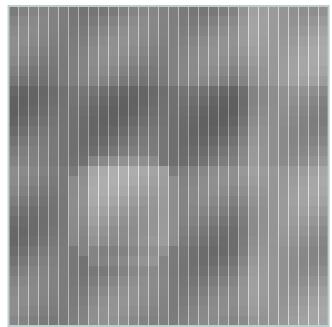
Blur



MSE 225 · PSNR 24.6 dB

SSIM 0.71

Blocking 8 × 8



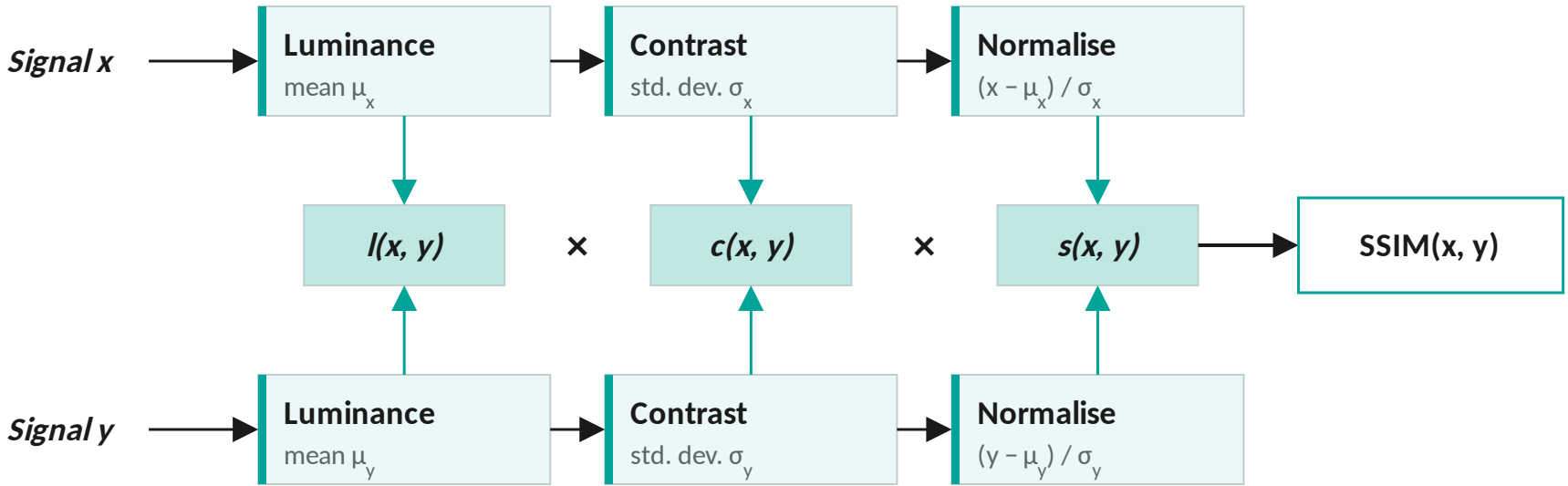
MSE 225 · PSNR 24.6 dB

SSIM 0.76

PSNR cannot tell these apart; SSIM, which compares local structure, can.

MSE treats every pixel error alike, regardless of **where** it is and **what structure** it destroys. A brightness shift barely changes the picture; blur and blocking destroy edges and texture — at identical PSNR. This is the classic argument of Wang & Bovik, *"Mean squared error: love it or leave it?"* (2009).

SSIM — the idea



Each comparison uses one statistic of both signals; their product is SSIM.
After Z. Wang et al., IEEE Trans. Image Processing 13(4), 2004, Fig. 3.

"The human visual system is highly adapted to extract structural information from a scene" — so measure the loss of **structure**, separately from changes of mean brightness and contrast, which the eye largely discounts (Wang, Bovik, Sheikh & Simoncelli, 2004).

SSIM — the formula

Local statistics of windows x, y (mean, std. dev., covariance):

$$\mu_x = \frac{1}{N}\sum_{i=1}^{N} x_i, \quad \sigma_x = \left(\frac{1}{N-1}\sum_{i=1}^{N}(x_i-\mu_x)^2\right)^{1/2}$$

$$l = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1}, \quad c = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}$$

$$s = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3}$$

With $C_3 = C_2/2$ the product $l \cdot c \cdot s$ simplifies to

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

- $C_1 = (K_1 L)^2$, $C_2 = (K_2 L)^2$, with $K_1 = 0.01$, $K_2 = 0.03$ and L the dynamic range — they keep the fractions stable when the denominators approach zero.
- The structure term compares the **normalised** signals $(x - \mu_x)/\sigma_x$ — it is their correlation coefficient.
- Evaluated in a sliding **11 × 11 Gaussian window** ($\sigma = 1.5$ px); the map is averaged into one **mean SSIM**.
- Range $[-1, 1]$; 1 only for identical windows.

MS-SSIM (Wang, Simoncelli & Bovik, 2003) evaluates contrast and structure at five scales, which accounts for viewing distance and resolution.

Worked example — one 3 × 3 window

A bright pixel on a flat background, and a **lower-contrast** copy of it ($L = 255$, $C_1 = 6.5$, $C_2 = 58.5$):

$$x = \begin{bmatrix} 60 & 60 & 60 \\ 60 & 150 & 60 \\ 60 & 60 & 60 \end{bmatrix} \qquad y = \begin{bmatrix} 65 & 65 & 65 \\ 65 & 110 & 65 \\ 65 & 65 & 65 \end{bmatrix}$$

1 · MSE and PSNR

$$\mathrm{MSE} = \tfrac{1}{9}\left(8 \cdot 5^2 + 40^2\right) = 200$$
$$\mathrm{PSNR} = 10 \log_{10} \tfrac{255^2}{200} \approx 25.1 \text{ dB}$$

2 · Statistics (with $1/(N-1)$, $N = 9$)

$$\mu_x = \mu_y = 70$$
$$\sigma_x^2 = \tfrac{8 \cdot 10^2 + 80^2}{8} = 900, \quad \sigma_y^2 = \tfrac{8 \cdot 5^2 + 40^2}{8} = 225$$
$$\sigma_{xy} = \tfrac{8 \cdot (-10)(-5) + 80 \cdot 40}{8} = 450$$

3 · The three terms — $l = 1$ (equal means), $s = 1$ ($\sigma_{xy} = \sigma_x \sigma_y$), and

$$c = \frac{2 \cdot 30 \cdot 15 + 58.5}{900 + 225 + 58.5} \approx 0.81$$

4 · Result: $\mathrm{SSIM} = l \cdot c \cdot s \approx \mathbf{0.81}$ — the structure is intact, only the contrast is halved.

A **uniform brightness shift** with the same MSE ($y = x + 14.1$) gives $c = s = 1$, $l \approx 0.98$: SSIM **0.98** at the same PSNR.

Try it — six distortions, one MSE



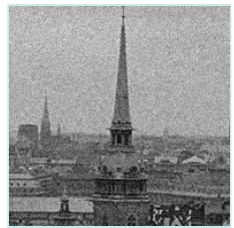
Reference



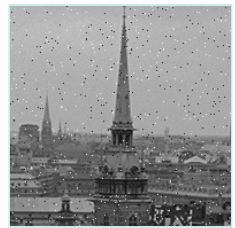
Mean shift
SSIM **0.99**



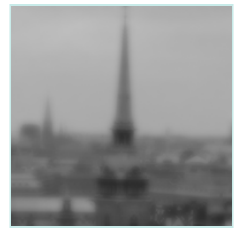
Contrast reduction
SSIM **0.96**



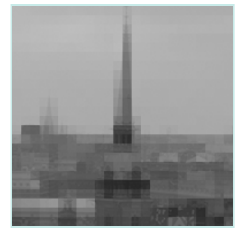
Gaussian noise
SSIM **0.52**



Impulse noise (2 %)
SSIM **0.70**



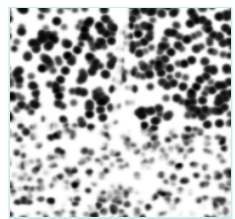
Blur
SSIM **0.67**



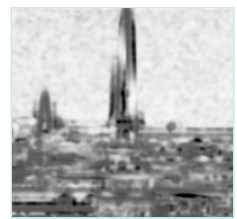
Blocking 8 × 8
SSIM **0.70**

SSIM map ↓
white = same structure
black = none





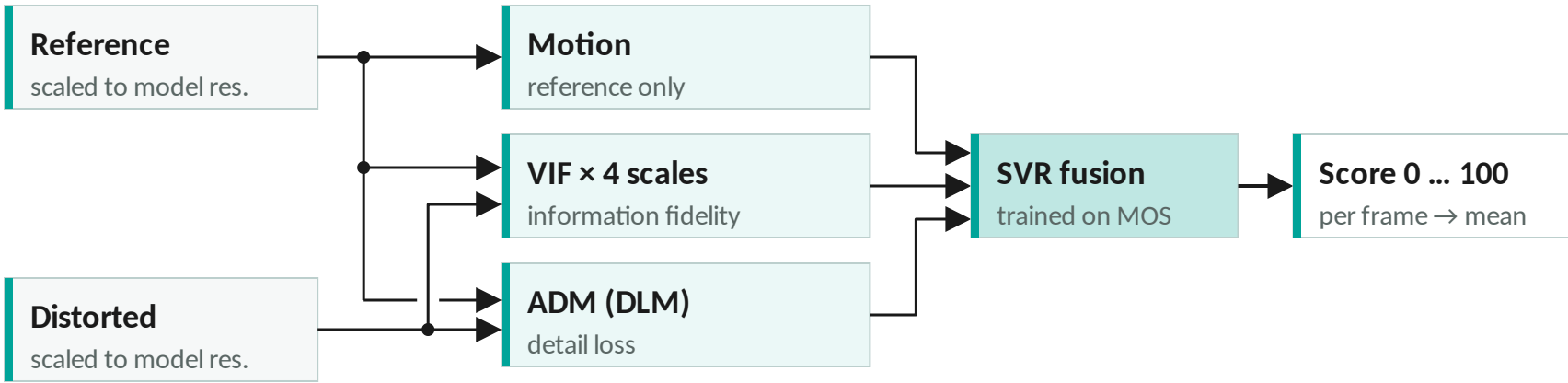




All six distortions are scaled to the **same MSE = 200**, so PSNR is identical for every one (25.1 dB). SSIM compares mean, contrast and structure in an 11 × 11 window around every pixel and ranks them very differently:

- **Mean shift (0.99):** Only the mean changes; contrast and structure are intact.
- **Gaussian noise (0.52):** New random structure everywhere; worst in the flat sky, masked in the busy city.
- **Blur (0.67):** Edges and fine texture removed; the flat sky has nothing to lose.
- **Contrast reduction (0.96):** Edges are weaker but all still there — only the contrast term drops.
- **Impulse noise (2 %) (0.70):** A few large hits; every pixel between them is exact.
- **Blocking 8 × 8 (0.70):** Texture inside each block lost, false edges added at block borders.

VMAF — Video Multimethod Assessment Fusion



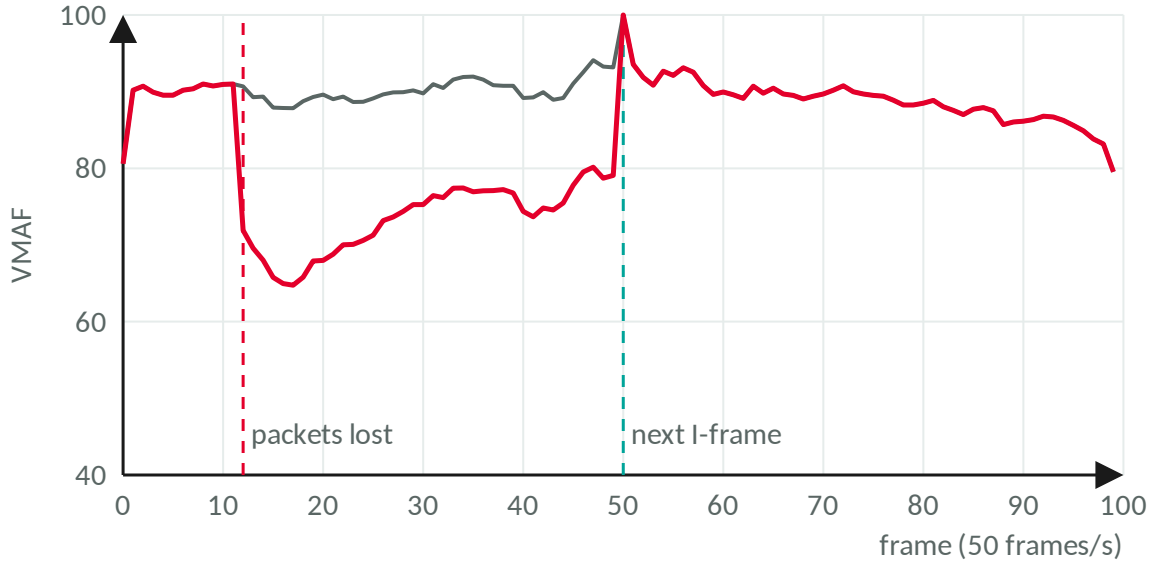
Models: vmaf_v0.6.1 (1080p TV at 3H) · 4K model (1.5H) · phone model · NEG variant (no enhancement gain).
The features are per-frame; temporal behaviour enters only through the motion feature and the pooling.

VIF — Visual Information Fidelity (Sheikh & Bovik, 2006): information shared by reference and distorted image, at four scales.

ADM / DLM — Detail Loss Metric (Li et al., 2011): lost detail, separated from additive impairments that distract attention.

Motion — mean co-located pixel difference of consecutive reference frames: artefacts are masked in fast motion.

From frames to one number — temporal pooling



error-free · mean 89.4

one loss burst

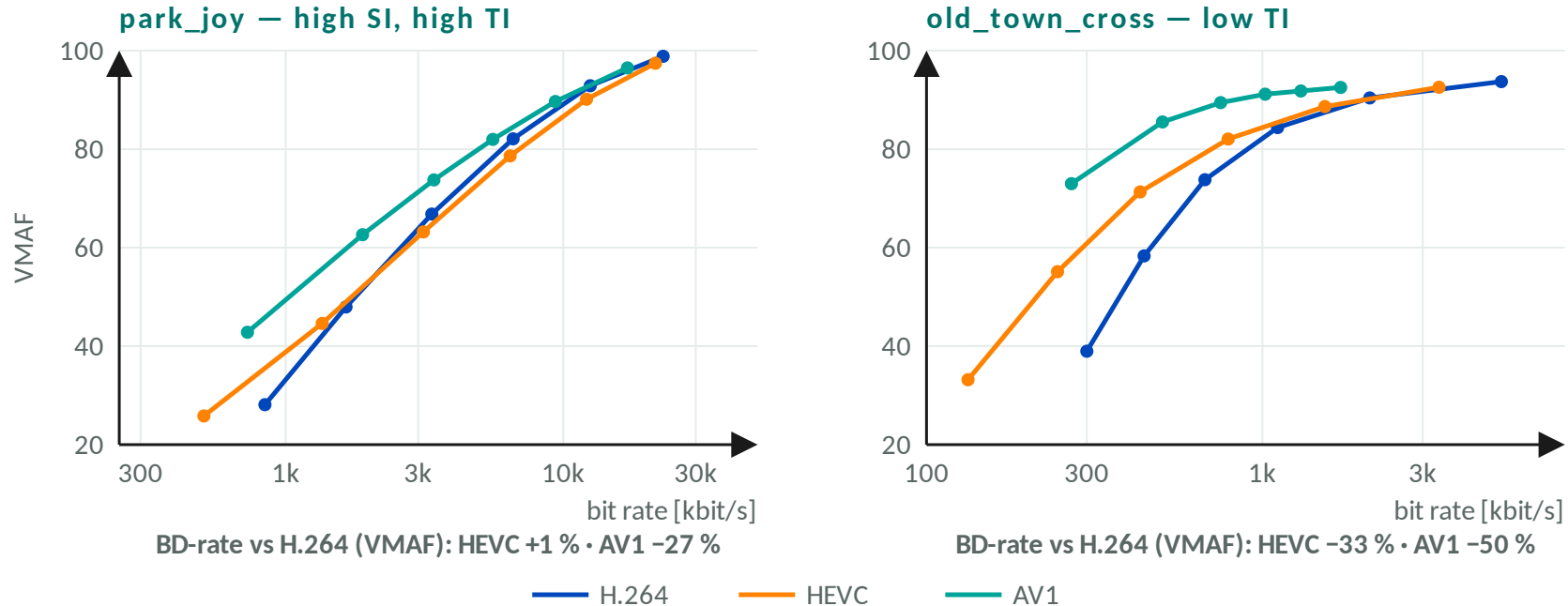
LOSSY CLIP, POOLED

arithmetic mean	**83.1**
harmonic mean	**82.2**
5th percentile	**68.0**
minimum	**64.8**

The same frames give a score between 65 and 83 — the pooling rule is part of the metric.

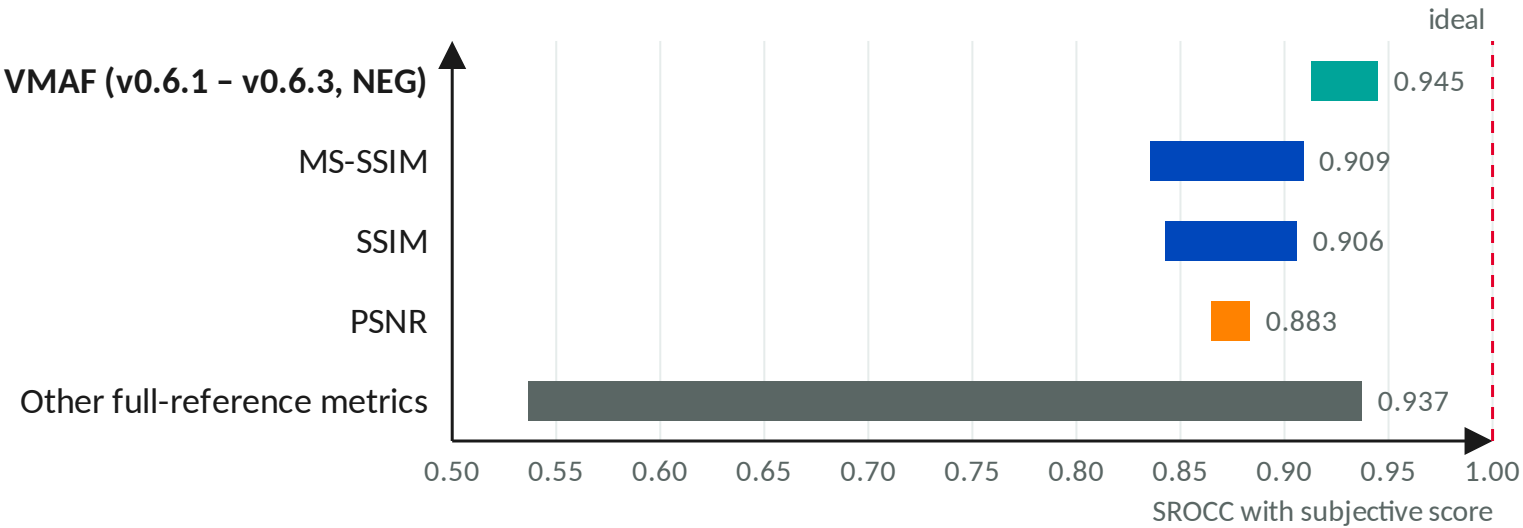
Every image metric produces **one value per frame**; a video needs one per clip. The **arithmetic mean** dilutes a short, severe drop; the **harmonic mean** (a VMAF option) and **low percentiles** weight the worst frames more, as viewers do. No pooling rule models **recency** — the last seconds of a clip weigh most in the viewer's memory.

Comparing codecs — rate–quality curves and BD-rate



Codecs are compared by their **rate–quality curves**. The **Bjøntegaard delta rate** (BD-rate) is the average horizontal distance between two curves: the bit-rate change needed for the same quality. The saving is **not a property of the codec alone** — on the demanding *park_joy* HEVC gains almost nothing over H.264, on *old_town_cross* a third.

How good are the metrics?



Bars span all variants of a metric (colour planes, versions). Data: MSU Video Quality Metrics Benchmark.

Metrics are validated by correlating their output with MOS over a database: **SROCC** (Spearman, rank order — monotonicity) and **PLCC** (Pearson, after a nonlinear mapping — accuracy), plus the RMSE of the prediction.

On compression artefacts all established metrics are close — PSNR too. The differences grow on other distortions (noise, enhancement, transmission errors), and every metric is only validated for the content and distortions of its test database.

No-reference and learned models

Natural scene statistics

Undistorted natural images follow characteristic statistical regularities; distortions break them. **BRISQUE** (Mittal, Moorthy & Bovik, UT Austin, 2012) learns quality from these features; **NIQE** (2013) needs no subjective scores at all, only a model of pristine images.

Deep learning

Convolutional networks extract spatial features per frame, **recurrent** networks or temporal pooling model their evolution over time — e.g. VSFA (2019), DOVER (2023). Trained on large user-generated-content databases (KoNViD-1k, YouTube-UGC) where no pristine reference exists.

Strength: work on **any** received video, including user uploads. Weakness: they learn their training data — including its biases — and generalise poorly to unseen distortions; none is yet an ITU standard.

Parametric models for adaptive streaming — ITU-T P.1203

HTTP adaptive streaming (DASH, HLS) changes the dominant impairments: packet loss is hidden by TCP, and the viewer sees instead **initial loading delay**, **stalling** and **quality switching**.

ITU-T P.1203 (2017) estimates the MOS of a 1–5 min streaming session from:

- per-segment bit rate, resolution, frame rate, codec
- initial loading delay, stalling events and their positions
- in modes 0–3: from metadata only (mode 0) up to full bitstream parsing (mode 3)

P.1204 (2020) extends the approach to H.264, HEVC and VP9 up to 4K.

What the subjective studies show

- one stall of a few seconds hurts more than a much longer initial delay (Hoßfeld et al., 2012)
- the number and duration of stalls dominate QoE; frequent quality switches are also penalised
- a recent drop weighs more than an early one — recency

P.1203 is to streaming video what the E-model is to VoIP: quality from parameters, available in every player.

Checkpoint — sharpening the score

An encoder vendor adds a sharpening filter before encoding. VMAF (default model) rises from 85 to 92 while PSNR falls by 1 dB. Is the video better?

Predict first · discuss with a neighbour · then reveal

> **Not necessarily.** Sharpening moves the output *away* from the source (PSNR falls) but boosts the detail and contrast features VMAF rewards — VMAF can be **gamed** by enhancement. Moderate sharpening may even be preferred by viewers; strong sharpening produces halos. Only a subjective test decides. Netflix introduced **VMAF NEG** (no enhancement gain) for exactly this case: codec comparisons should report it alongside the default model.

Common mix-ups

Codec = container

MP4, MKV and TS are containers; H.264, HEVC and AV1 are codecs. The file name names only the container.

4K = twice Full HD

Both dimensions double: four times the pixels, four times the raw bit rate.

Bit rate = quality

The same bit rate gives very different quality on different content (SI/TI). Compare codecs on a range of contents.

MAX in PSNR

It is the maximum *possible* value (255, 1023), not the brightest pixel of the image.

Averaging PSNR

Mean of frame PSNRs ≠ PSNR of the mean MSE. Always state the pooling.

SSIM is not a percentage

SSIM 0.95 is not "95 % quality"; its scale is nonlinear and content-dependent.

VMAF is not no-reference

VMAF needs the source video, exactly like PSNR and SSIM.

MOS vs. DMOS

ACR gives MOS; ACR-HR and DCR give DMOS. Different scales, not interchangeable.

Summary — putting a number on video quality

Raw video is huge $W \cdot H \cdot f \cdot b_{\text{px}}$: 746 Mbit/s for 1080p30 in Y′CbCr 4:2:0. Codecs compress it 150–200 : 1; containers only carry the result.

Quality is lost along the whole chain Hybrid coding trades bits for artefacts through quantisation; inter prediction makes a single loss propagate to the next I-frame.

Subjective tests are the reference P.910 and BT.500 fix viewers, content (SI/TI), conditions and method — ACR, ACR-HR, DCR, PC, DSCQS — and yield MOS or DMOS with confidence intervals.

Full-reference metrics compare with the source PSNR measures error energy; SSIM compares luminance, contrast and structure; VMAF fuses VIF, ADM and motion with a model trained on MOS.

Deployable models need less input No-reference models judge the received video alone; parametric models such as P.1203 predict streaming QoE from bit rate, resolution and stalling.

Exit questions — use the ideas

1. Why does a packet loss in a P-frame degrade quality for longer than the same loss in a B-frame?
2. Why must a subjective test state the viewing distance, and what would you expect to change if viewers moved from 3 H to 6 H?
3. Two degraded frames have the same PSNR. Construct a distortion for which SSIM stays close to 1, and one for which it drops sharply.
4. A network operator wants to monitor video quality on live IPTV traffic. Which families of models can it use, and which can it not?
5. What is the raw bit rate of 4K at 50 frames/s, 10-bit 4:2:0 — and what compression ratio does a 20 Mbit/s stream of it need?

Explain the mechanism, not just the name.

Where you use this — Lab 04

Lab 04 applies the full-reference part of this lecture to still images:

- implement **MSE and PSNR** yourself, then compare with the reference implementation
- compute **SSIM** with the supplied loop and vectorised versions, and inspect the **SSIM map**
- compare **dense and sparse** noise at similar MSE — and rank the images by PSNR, by SSIM and by your own eye
- vary the SSIM **window** (uniform vs Gaussian, 7 to 15 px)

A video metric is an image metric per frame plus **temporal pooling** — everything in the lab carries over frame by frame.

```
ssh <lab-server>
cd qos-04 && uv sync
uv run jupyter lab --ip 0.0.0.0
```

References — standards

- ITU-T P.910, *Subjective video quality assessment methods for multimedia applications*, 1996; editions 1999, 2008, 2021–2023; current edition 2026.
- ITU-R BT.500-15, *Methodologies for the subjective assessment of the quality of television images*, 2023.
- ITU-R BT.1788, *Methodology for the subjective assessment of video quality in multimedia applications* (SAMVIQ), 2007.
- ITU-T P.913, *Methods for the subjective assessment of video quality, audio quality and audiovisual quality of Internet video and distribution quality television in any environment*, 2021.
- ITU-T P.1203, *Parametric bitstream-based quality assessment of progressive download and adaptive audiovisual streaming services over reliable transport*, 2017. ITU-T P.1204, 2020.
- ITU-R BT.601-7 (SDTV), 2011; BT.709-6 (HDTV), 2015; BT.2020-2 (UHDTV), 2015 — colour spaces and sampling. ISO/IEC 14496-12 (ISO BMFF), ISO/IEC 13818-1 (MPEG-TS).
- ITU-T H.264 (AVC), 2003; H.265 (HEVC), 2013; H.266 (VVC), 2020. AOMedia, *AV1 Bitstream & Decoding Process Specification*, 2019.

References — literature

- Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Trans. Image Processing*, 13(4), 600–612, 2004.
- Z. Wang, A. C. Bovik, "Mean squared error: love it or leave it?," *IEEE Signal Processing Magazine*, 26(1), 98–117, 2009.
- H. R. Sheikh, A. C. Bovik, "Image information and visual quality," *IEEE Trans. Image Processing*, 15(2), 430–444, 2006.
- Z. Li, A. Aaron, I. Katsavounidis, A. Moorthy, M. Manohara, "Toward a practical perceptual video quality metric," Netflix Technology Blog, 2016; github.com/Netflix/vmaf.
- A. Mittal, A. K. Moorthy, A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Trans. Image Processing*, 21(12), 4695–4708, 2012.
- M. Seufert et al., "A survey on quality of experience of HTTP adaptive streaming," *IEEE Communications Surveys & Tutorials*, 17(1), 469–492, 2015.
- A. Antsiferova et al., "Video compression dataset and benchmark of learning-based video-quality metrics," *NeurIPS Datasets and Benchmarks*, 2022.

Thank you for your attention

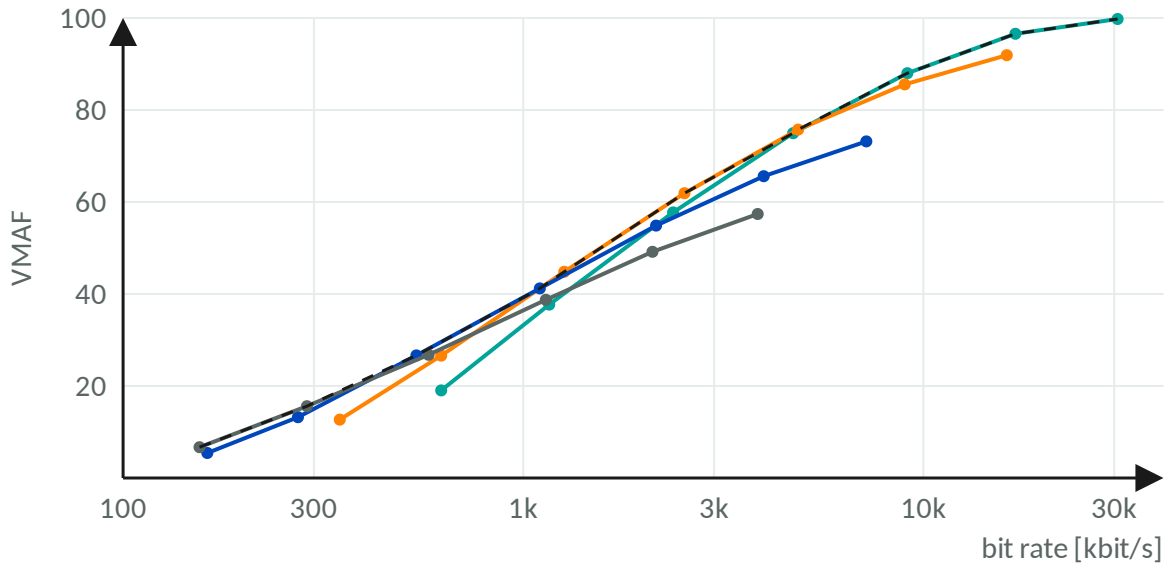
Jan Rozhon

+420 596 995 900 · jan.rozhon@vsb.cz

Extras

Beyond the standard 2D picture

Extras — per-title encoding and the convex hull



720p
540p
360p
270p
convex hull (best of all)

FIRST ON THE HULL

270p	from 155 kbit/s
360p	from 541 kbit/s
540p	from 1 266 kbit/s
720p	from 4 726 kbit/s

park_joy, H.264 medium, CRF 18–42. Lower resolutions are upscaled (bicubic) to 1280 × 720 before VMAF.

An adaptive-streaming service offers each title at several **resolution–bit-rate pairs** (the *ladder*). Below a few Mbit/s, a *lower* resolution upscaled by the player looks better than a starved full-resolution encode. The **convex hull** over all encodes gives the best ladder for *this* clip; a fixed ladder for all titles wastes bits on easy content and starves hard content (Netflix, *per-title encoding*, 2015).

Extras — HDR, wide colour gamut and 360° video

High dynamic range, wide colour gamut (ITU-R BT.2100)

Pixel values are no longer proportional to perceived brightness in the way 8-bit SDR assumes: PQ and HLG transfer functions, 10-bit samples, peak luminance up to 1 000 cd/m² and more. PSNR and SSIM on code values mis-weight errors in dark and bright regions. Used instead: metrics in perceptual colour spaces (**ΔE ITP**, ITU-R BT.2124), **HDR-VDP** (Mantiuk et al.), and HDR-specific VMAF models.

360° video (ITU-T P.919 for subjective tests)

The sphere is stored in a flat projection (equirectangular, cube map), which over-represents the poles. **WS-PSNR** weights each pixel by the solid angle it covers; only the **viewport** the viewer actually looks at matters, and head-motion data decide which. Subjective tests use head-mounted displays and must control simulator sickness.

Extras — crowdsourced subjective tests

Laboratory tests are slow and expensive; **crowdsourcing** runs the same ACR, ACR-HR, DCR or CCR test on hundreds of remote workers in hours.

What changes

- no control of display, distance, light or attention
- far more votes per condition compensate for the extra noise
- quality control becomes part of the method: qualification tests, **gold questions** with known answers, **trapping questions**, environment and hardware checks

Standards and practice

- ITU-T **P.913** — tests outside the laboratory, including crowdsourcing
- ITU-T **P.808** — the crowdsourcing counterpart for speech (Lecture 05)
- open-source P.910 crowdsourcing toolkit (Naderi & Cutler, 2022) — shown to reproduce laboratory results for ACR, ACR-HR, DCR and CCR

Crowdsourced MOS is what most learned no-reference models are trained on — their bias starts here.